

Video Refresher — Amii × ZeroKey Technical Interview

Job ID: 1085 · **Companion to:** `amii-zerokey-ml-resident-technical.md` (section numbers below match it) and `amii-zerokey-case-scenarios.md` **Compiled:** 2026-09-30

Every link below was found through a web search that returned the video's own YouTube page. Videos are chosen for **refreshing**, not learning from scratch: short, visual explainers first, with a longer lecture only where the topic needs it.

Channels you'll see most: – **StatQuest** (Josh Starmer): statistics and classical ML, 5–20 min each. – **3Blue1Brown**: visual intuition for maths and deep learning. – **UMich EECS 498** (Justin Johnson): full computer-vision lectures, ~1 hour each; skim at 1.5×. – **First Principles of Computer Vision** (Shree Nayar, Columbia): short, precise geometry and optical-flow lectures. – **Cyrill Stachniss** and **MATLAB**: Kalman filtering and state estimation.

If you only have ~5 hours

Watch these in order. They cover the Tier 1 topics from the study-priority table.

#	Video	Why
1	Machine Learning Fundamentals: Bias and Variance (StatQuest)	Fundamentals warm-up
2	p-values: What they are and how to interpret them (StatQuest)	The most commonly botched stats question
3	ROC and AUC, Clearly Explained! (StatQuest)	Evaluation under imbalance
4	Long Short-Term Memory (LSTM), Clearly Explained (StatQuest)	Your core model family
5	Attention in transformers, visually explained (3Blue1Brown)	Attention, $\sqrt{d_k}$, Q/K/V
6	Why Use Kalman Filters? — Understanding Kalman Filters, Part 1 (MATLAB)	Headline motion topic
7	Kalman Filter & EKF (Cyrill Stachniss)	The equations, derived properly
8	Dynamic Time Warping (DTW) Explained	Your DTW → RNN story
9	Lecture 16: Detection and Segmentation (UMich EECS 498)	Detection, Mask R-CNN, keypoints
10	Lecture 18: Videos (UMich EECS 498)	Two-stream, 3D CNNs, SlowFast
11	ML System Design Question — Create an ETA System for Maps (full mock)	See the case-study format done live

§1.1 Core ML concepts

Concept	Video	Focus on
Bias–variance	Machine Learning Fundamentals: Bias and Variance (StatQuest)	Overfitting vs. underfitting intuition
Cross-validation	Machine Learning Fundamentals: Cross Validation (StatQuest)	k-fold; then think "why this breaks for time series"

Concept	Video	Focus on
L2 regularization	Regularization Part 1: Ridge (L2) Regression (StatQuest)	Shrinkage
L1 regularization	Regularization Part 2: Lasso (L1) Regression (StatQuest)	Sparsity
Why L1 gives zeros	Ridge vs Lasso Regression, Visualized!!! (StatQuest)	The geometric picture
Neural networks	But what is a neural network? (3Blue1Brown)	Only if you want a warm-up
Gradient descent	Gradient descent, how neural networks learn (3Blue1Brown)	Loss landscape intuition
Backpropagation	Backpropagation, intuitively and Backpropagation calculus (3Blue1Brown)	Chain rule; vanishing gradients
Adam	Adam Optimization Algorithm (C2W2L08) (Andrew Ng)	Momentum + RMSprop combined
Random forest	Random Forests Part 1 — Building, Using and Evaluating (StatQuest)	Bagging reduces variance
Gradient boosting	Gradient Boost Part 1: Regression Main Ideas (StatQuest)	Fitting residuals sequentially
XGBoost	XGBoost Part 1: Regression (StatQuest)	Regularized boosting
PCA	Principal Component Analysis (PCA), Step-by-Step (StatQuest)	Eigenvectors of the covariance matrix
SVD	Singular Value Decomposition (SVD): Overview (Steve Brunton)	Also needed for Kabsch (§6.7)
Entropy, cross-entropy, KL	A Short Introduction to Entropy, Cross-Entropy and KL-Divergence	Why cross-entropy is the classification loss

§1.2 Statistics

Concept	Video	Focus on
Bayes' rule	Bayes theorem, the geometry of changing beliefs (3Blue1Brown)	Base-rate fallacy: directly relevant to rare-event alarms
MLE	Maximum Likelihood, clearly explained!!! (StatQuest)	Then link MLE → MSE/cross-entropy losses
Probability vs. likelihood	In Statistics, Probability is not Likelihood. (StatQuest)	A common trick question
Central limit theorem	But what is the Central Limit Theorem? (3Blue1Brown)	Why $SE = \sigma/\sqrt{n}$
Standard error	The standard error, Clearly Explained!!! (StatQuest)	
Confidence intervals	Confidence Intervals, Clearly Explained!!! (StatQuest)	Frequentist interpretation
Bootstrap	Bootstrapping Main Ideas!!! (StatQuest)	Then remember: block bootstrap for time series
p-values	p-values: What they are and how to interpret them (StatQuest)	What a p-value is <i>not</i>
Multiple comparisons	False Discovery Rates, FDR, clearly explained (StatQuest)	Benjamini–Hochberg

Concept	Video	Focus on
Simpson's paradox	Simpson's Paradox (MinutePhysics)	Confounders (shift, experience)
Difference-in-differences	Causality: Difference-in-Differences	"Did the intervention reduce risk?"
Stationarity, ACF/PACF, ARIMA	Understanding Stationarity, ACF, PACF, and ARIMA Fundamentals	Classical time-series baseline
ROC vs. PR	ROC and AUC, Clearly Explained! (StatQuest) and Precision-Recall Curve Explained with Simple Examples	Why PR-AUC under imbalance
Calibration	Probability Calibration for Classification (Platt, isotonic, logistic and beta)	Reliability diagrams, Platt scaling

§2 Data handling and evaluation

Concept	Video	Focus on
Time-series CV / leakage	Time Series Cross Validation Explained	Forward-chaining splits
Data leakage in general	Understanding and Avoiding Data Leakage with Hamel Husain	Practical leakage stories; pairs with Scenario T

§3 Deep learning for time series

Concept	Video	Focus on
RNNs	Lecture 12: Recurrent Neural Networks (UMich EECS 498)	Only the vanishing-gradient part if short on time
LSTM	Long Short-Term Memory (LSTM), Clearly Explained (StatQuest)	Gates and the cell-state path
TCN	Temporal Convolutional Networks, The Next Revolution for Time-Series?	Dilated causal convolutions; it includes a motion-detection case study
Attention	Attention for Neural Networks, Clearly Explained!!! (StatQuest) and Attention in transformers, visually explained (3Blue1Brown)	Q/K/V, scaling, multi-head
Transformers	Transformer Neural Networks, ChatGPT's foundation, Clearly Explained!!! (StatQuest)	Positional encoding, the block
Graph networks	Lecture 6.1 — Introduction to Graph Neural Networks (Stanford CS224W)	Message passing
ST-GCN	ST-GCN: Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition	Closest model to RTLS multi-tag data
Temporal segmentation	MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation (CVPR 2019)	SOP step segmentation
HMM / Viterbi	Hidden Markov Models (HMMs) Explained with the Viterbi Algorithm	Steps as hidden states

Concept	Video	Focus on
Self-supervised / contrastive	Tutorial 17: Self Supervised Contrastive Learning with SimCLR (Part 1)	The idea transfers to TS2Vec
Time-series foundation models	Chronos: Learning the Language of Time Series	Tokenizing values; zero-shot forecasting

No good video found for PatchTST. Read the abstract of the paper ("*A Time Series is Worth 64 Words*", Nie et al, 2023) instead.

§4 Deep learning for computer vision

Concept	Video	Focus on
CNNs	Lecture 7: Convolutional Networks (UMich EECS 498)	Receptive fields, weight sharing
ResNet and architectures	Lecture 8: CNN Architectures (UMich EECS 498)	Residual connections
Vision Transformer	An Image is Worth 16x16 Words (Paper Explained) (Yannic Kilcher)	Patches as tokens; needs pre-training
DINO	DINO: Emerging Properties in Self-Supervised Vision Transformers (Yannic Kilcher)	Label-free features
SAM	Segment Anything Paper Explained	Promptable segmentation
Object detection	Lecture 15: Object Detection (UMich EECS 498)	IoU, NMS, one- vs. two-stage
Segmentation, Mask R-CNN, keypoints	Lecture 16: Detection and Segmentation (UMich EECS 498)	Includes keypoint (pose) R-CNN
DETR	DETR — End to end object detection with transformers (ECCV 2020)	Set prediction, no NMS
U-Net	Semantic Segmentation with U Net Made Easy	Encoder-decoder with skips
Pose estimation	Realtime Multi-Person 2D Human Pose Estimation using Part Affinity Fields, CVPR 2017 Oral	Bottom-up pose (OpenPose)
Video / action recognition	Lecture 18: Videos (UMich EECS 498)	Two-stream, I3D, SlowFast
Optical flow	Overview — Optical Flow and Lucas-Kanade Method (First Principles of CV)	Brightness constancy
Multi-object tracking	DeepSORT Object Tracking Explained	Kalman + Hungarian matching + appearance
Pinhole camera, $K[R t]$	Linear Camera Model and Intrinsic and Extrinsic Matrices (First Principles of CV)	
Homography	Computing Homography and Dealing with Outliers: RANSAC (First Principles of CV)	Camera → floor-plan mapping
Epipolar geometry	Epipolar Geometry — Uncalibrated Stereo (First Principles of CV)	Multi-view triangulation
Industrial anomaly detection (PatchCore)	Paper review: Towards Total Recall in Industrial Anomaly Detection	Memory bank of patch features

§5 Practical industry tasks and deployment

Concept	Video	Focus on
Quantization	From FP32 to INT8: Post-Training Quantization Explained in PyTorch	Scale/zero-point, PTQ vs. QAT
Knowledge distillation	Knowledge Distillation: A Good Teacher is Patient and Consistent	Teacher → edge student
Anomaly detection (classical)	Gaussian Mixture Model (GMM) for Anomaly Detection	Density-based baseline
Change-point detection	Change Point Detection in Time Series and Bayesian Online Change-Point Detection	Segmenting streams (Scenario I)

§6 Mathematical methods for analyzing motion

Concept	Video	Focus on
Kalman intuition	Why Use Kalman Filters? — Understanding Kalman Filters, Part 1 (MATLAB)	Continue with the rest of the MATLAB series if time allows
Kalman in 5 minutes	Kalman Filter — 5 Minutes with Cyrill	Quick recap the night before
Kalman + EKF equations	Kalman Filter & EKF (Cyrill Stachniss)	Predict/update, linearization
UKF	SLAM Course — 06 — Unscented Kalman Filter (Cyrill Stachniss)	Sigma points
Particle filter	MSR Course — 07 Particle Filter (Cyrill Stachniss)	Multimodal cases
DTW	Dynamic Time Warping (DTW) Explained	Dynamic-programming table
Fourier transform	But what is the Fourier Transform? A visual introduction (3Blue1Brown)	Repetitive-motion analysis
Quaternions	Quaternions and 3d rotation, explained interactively (3Blue1Brown)	q and -q are the same rotation
Euler angles / gimbal lock	Euler Angles and Gimbal Lock, Explained	Why not Euler angles
Kabsch / Procrustes	Kabsch-Umeyama Algorithm — How to Align Point Patterns	Rotation via SVD (Scenario P)
Multilateration	How Does Trilateration Work in GPS Positioning?	Same maths as ultrasonic ranging
NIOSH lifting equation	The NIOSH Lifting Equation: Example 2	Worked RWL example
PINNs	Physics Informed Neural Networks (PINNs) (Steve Brunton)	Xu Han's area
Neural ODEs	Neural Ordinary Differential Equations (Neural ODEs): Implementation	ResNet ↔ Euler step

No good videos found for Savitzky–Golay filtering, the RTS smoother, Fréchet distance, or KalmanNet. The prep guide's own sections (§6.2–§6.4) are enough for the interview.

§7 AI agents

Concept	Video	Focus on
LLM overview	1hr Talk: Intro to Large Language Models (Andrej Karpathy)	Tools, agents, security sections
ReAct	ReAct AI Agents, clearly explained!	Reason → act → observe loop
MCP	Model Context Protocol (MCP), clearly explained (why it matters)	Tools, resources, prompts
RAG	What is Retrieval-Augmented Generation (RAG)? (IBM Technology)	Grounding answers in SOP documents
LoRA	Low-Rank Adaptation (LoRA) Explained	You used it on MusicGen

§8 Reinforcement learning (just in case)

Concept	Video	Focus on
RL foundations	RL Course by David Silver — Lecture 1: Introduction to Reinforcement Learning	MDPs, value functions; Amii's home field
RL in one lecture	Lecture 21: Reinforcement Learning (UMich EECS 498)	Q-learning, policy gradients, faster than Silver
PPO	Proximal Policy Optimization Explained	The clipped objective

Part B — Case study practice

Video	Why
ML System Design Question — Create an ETA System for Maps (full mock)	Closest to a time-series/industry problem
ML System Design Mock Interview — Build an ML System That Classifies Which Tweets Are Toxic	Rare-positive classification, like risk detection
Full ML Design Mock by ex-Meta Staff Engineer (with feedback)	The feedback section shows what interviewers reward

These mocks are recommendation- and web-flavoured, so watch them for **structure and pacing**, not content. Your case will be industrial.

Suggested schedule (if the interview is 2–3 days out)

- **Day 1 (~3 h):** the "~5 hours" list, items 1–8.
- **Day 2 (~3 h):** items 9–11, then §4 pose, tracking and camera geometry; §6 quaternions and Kabsch.
- **Day 3 (~1.5 h):** §7 agents (ReAct, MCP), one case-study mock, then rehearse Scenarios T and I out loud.